Retrieval-Ready Knowledge: Chunking and Metadata

DSBA 6171 | Evidence-based observations from saved n8n chunk exports

Scope: Five documents, two chunking configurations, using the uploaded baseline_A_chunks.json (69 records) and alternative_B_chunks.json (144 records). Findings describe the saved exports, not a completed retrieval-quality benchmark.

1. Experiment configuration and output

Metric	Configuration A	Configuration B
Chunk size	1,600 characters	800 characters
Chunk overlap	200 characters	100 characters
Documents	5	5
Saved chunks	69	144
Embedding model	nomic-embed-text	nomic-embed-text

Configuration B created 75 more chunks (approximately 108.7% more) than Configuration A. This is an output-volume result, not evidence that B has higher retrieval accuracy.

2. Per-document chunk counts

Document / version	A	B	Authority
CFPB Circular 2023-03	16	33	CURRENT
Loan Approval Escalation Procedure	27	57	CURRENT
2024 Underwriting Policy (superseded)	8	17	SUPERSEDED
2026 Revised Underwriting Policy	13	27	CURRENT
Superseded Underwriting Policy v2	5	10	SUPERSEDED
TOTAL	69	144	

3. Representative chunk observations

Example 1 - CFPB Circular 2023-03 (DOC001, v1.0, CURRENT). Baseline A chunk index 0 contains both the question about whether creditors can rely on generic CFPB sample reasons and the response beginning "No, creditors may not rely...". Alternative B chunk index 0 ends at the "Response" heading; its answer appears in the next chunk. This illustrates the risk of splitting a question from its answer when using smaller chunks.

Example 2 - 2024 underwriting policy (FIN-POL-001, v1, SUPERSEDED). The historical policy is explicitly tagged SUPERSEDED in both exports. The combination of document ID, version, and authority status distinguishes it from later versions.

Example 3 - 2026 revised underwriting policy (FIN-POL-001, v3, CURRENT). The revised policy is tagged CURRENT in both exports. A retrieval system should use these attributes to prioritize the applicable version when asked about current underwriting requirements.

4. Metadata and provenance observations

Each saved record includes chunk_id, doc_id, source_path, version, authority_status, effective_date, chunk_index, start_char, chunk_text, and metadata_json. Embedded metadata_json fields include title, document type, source, owner, jurisdiction/region, confidentiality, and access group where supplied.

Versioning: FIN-POL-001 is reused across v1, v2, and v3. Document ID alone is therefore insufficient to identify a unique policy version; use doc_id + version and retain authority status and effective date.

Provenance cleanup: The CFPB metadata source URL in both uploaded exports still includes the tracking suffix `?utm_source=chatgpt.com` (16 A chunks and 33 B chunks). The GitHub metadata may already have been corrected, but these particular exports still reflect the older URL. This does not affect chunk counts; refresh exports if the final evidence must show the clean URL.

5. Interpretation and limitations

Configuration A provides longer contiguous context, which may help keep related explanations together. Configuration B creates smaller, more focused passages, which may help isolate a relevant statement but can separate adjacent reasoning. These are plausible trade-offs illustrated by the exports, not measured retrieval outcomes.

The files demonstrate chunk creation and metadata preservation. They do not include ranked search results, query embeddings, retrieval scores, answer correctness, or evidence that superseded content is filtered at query time. Do not claim improved retrieval accuracy or authority enforcement until those behaviors are tested.

6. Recommended retrieval evaluation

Use identical source documents, the same embedding model and Top-K (suggested: 5), and the same questions for both configurations. Suggested questions: (1) What reasons must creditors provide when AI leads to adverse action? (2) Which underwriting policy is currently authoritative? (3) What is the loan-approval escalation procedure? Record top retrieved chunk IDs, document/version/authority, relevance, whether the answer is complete, and whether superseded material is incorrectly treated as current.

7. n8n workflow check

The known workflow nodes are Manual Trigger, List GitHub Documents, Build Document Manifest, Ingest and Embed Documents, Preview Chunk Metadata, and Format Chunk Metadata. These perform ingestion and inspection, not necessarily query-based retrieval. Inspect the canvas for any separate nodes named Search, Retrieve, Vector Search, Query, or HTTP Request calling a search endpoint. A screenshot of the complete canvas will establish whether retrieval is already implemented.

Evidence source: user-provided baseline_A_chunks.json and alternative_B_chunks.json. Configuration parameters were supplied in the experiment setup; chunk counts and metadata findings were verified from the exported files.